

MYS MENSHIKOVA, Russian Federation

WMO: 209430

Lat: 70.719N

Lon: 57.563E

Elev: 14

StdP: 101.16

Time Zone: 3.00 (E03)

Period: 82-94

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.9	-30.8	-35.5	0.1	-32.6	-33.6	0.2	-30.8	16.5	-5.7	14.9	-8.3	4.3	340	1.102

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	2.7	11.2	10.0	9.6	8.5	8.5	7.6	10.2	11.1	8.7	9.4	7.7	8.3	4.8	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.5	7.4	10.6	8.2	6.8	9.2	7.1	6.3	8.1	29.8	11.1	26.5	9.5	24.2	8.5	16.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.0	14.2	12.7	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-6.3	-18.5	-17.4	-13.8	-12.6	-4.9	1.1	4.5	4.7	3.8	-2.4	-7.4	-12.8
	DBStd	10.02	8.25	8.71	7.47	6.88	4.15	2.73	2.83	2.90	1.71	4.09	5.94	6.57
	HDD10.0	5939	885	766	739	679	462	268	173	166	187	383	523	708
	HDD18.3	8977	1143	999	998	929	720	518	429	422	437	642	773	966
	CDD10.0	4	0	0	0	0	0	0	2	2	0	0	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
	WSAvg	6.5	6.7	6.6	6.5	6.0	6.0	5.5	5.6	5.9	6.6	7.6	7.6	7.6
	PrecAvg	426	26	25	20	19	21	27	35	46	55	63	49	40
Precipitation	PrecMax	598	62	50	49	47	49	62	70	93	118	107	108	91
	PrecMin	177	3	1	1	1	3	11	3	23	14	29	10	7
	PrecStd	98	18	15	13	10	11	13	19	16	25	22	22	20
	PrecStd	98	18	15	13	10	11	13	19	16	25	22	22	20
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.9	0.7	0.5	1.1	2.2	10.5	15.9	13.0	8.7	5.6	1.3	-0.3
		MCWB	-1.5	0.0	-0.3	0.3	1.3	8.6	13.2	11.2	7.3	4.8	0.8	-0.9
	2%	DB	-1.9	-1.4	-0.4	0.4	1.2	7.7	11.6	11.3	7.8	4.8	0.7	-1.3
		MCWB	-2.6	-1.9	-0.9	-0.2	0.6	6.6	10.5	10.1	7.1	4.3	0.2	-2.0
	5%	DB	-3.3	-2.8	-1.4	-0.2	0.8	6.2	9.7	9.8	7.1	3.8	0.4	-2.1
		MCWB	-3.8	-3.3	-1.9	-0.7	0.2	5.0	8.6	8.8	6.6	3.4	0.0	-2.6
	10%	DB	-5.2	-4.2	-2.6	-1.4	0.2	4.7	8.6	8.7	6.4	2.7	-0.1	-3.2
		MCWB	-5.7	-4.7	-3.2	-1.9	-0.3	3.8	7.7	7.7	5.8	2.3	-0.4	-3.8
	0.4%	WB	-1.3	0.2	0.1	0.6	1.6	9.1	13.6	11.7	7.9	5.1	1.1	-0.8
		MCDB	-1.0	0.7	0.5	0.9	2.0	10.1	15.3	12.5	8.2	5.3	1.1	-0.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	-2.5	-1.8	-0.9	0.0	0.8	6.8	10.6	10.1	7.3	4.2	0.5	-1.9
		MCDB	-2.0	-1.5	-0.5	0.3	1.0	7.8	11.7	11.1	7.7	4.6	0.7	-1.5
	5%	WB	-3.8	-3.2	-1.9	-0.7	0.4	5.2	8.8	9.1	6.7	3.3	0.1	-2.7
		MCDB	-3.2	-2.7	-1.5	-0.3	0.7	5.9	9.4	9.8	7.0	3.7	0.4	-2.2
	10%	WB	-5.7	-4.7	-3.2	-1.9	-0.2	3.8	7.7	7.9	5.9	2.4	-0.4	-3.8
		MCDB	-5.1	-4.3	-2.7	-1.4	0.2	4.6	8.5	8.6	6.3	2.7	-0.1	-3.2
Mean Daily Temperature Range	5% DB	MDBR	6.6	6.8	6.9	7.2	4.4	3.0	3.3	2.7	2.3	4.0	4.6	6.4
		MCDBR	8.4	7.8	6.3	6.1	3.3	5.6	5.7	4.4	3.6	3.9	3.1	7.2
		MCWBR	8.6	7.9	6.4	6.1	3.1	4.5	4.6	3.4	3.2	3.7	3.0	7.2
	5% WB	MCDBR	8.5	7.8	6.4	6.0	3.0	5.4	5.4	4.4	3.0	3.4	3.0	6.8
		MCWBR	8.6	7.9	6.5	6.1	3.0	4.7	4.7	3.5	2.9	3.4	3.1	6.8
		MCWBR	8.6	7.9	6.5	6.1	3.0	4.7	4.7	3.5	2.9	3.4	3.1	6.8
Clear-Sky Solar Irradiance	taub		N/A	0.239	0.287	0.317	0.333	0.333	0.366	0.348	0.314	0.257	N/A	N/A
	taud		N/A	1.940	2.024	2.056	2.108	2.282	2.370	2.459	2.461	2.145	N/A	N/A
	Ebn at Noon		0	416	678	793	833	852	805	777	694	408	0	0
	Edh at Noon		0	41	83	113	123	107	96	78	57	32	0	0
All-Sky Solar Radiation	RadAvg		0.00	0.26	1.70	3.81	5.54	5.83	4.97	3.07	1.48	0.36	0.00	0.00
	RadStd		0.00	0.05	0.20	0.32	0.52	0.59	0.60	0.29	0.10	0.05	0.00	0.00

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	+1.72	+2.78	+3.36	+1.64	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page